# Alexandru-Georgian Oprea

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## Education

**ETH Zürich - Department of Physics** — *Zürich, Switzerland*

MSc Physics — *Sep 2025–Mar 2027*

- Master's program focused on theoretical physics, with additional mathematics coursework
- Current GPA: 5.57/6; 121 ECTS completed

**University of Groningen - Faculty of Science and Engineering** — *Groningen, Netherlands*

BSc Astronomy — *Sep 2022–Aug 2025*

- Graduated summa cum laude; completed 320 ECTS
- GPA: 9.1/10; ranked 2nd of 115 students

**University of Groningen** — *Groningen, Netherlands*

Honours College — *Apr 2023–Aug 2025*

- Selective extracurricular program for top students across faculties

## Publications & Research Experience

**Free Subalgebras of the Grothendieck–Teichmüller Lie Algebra (MSc thesis)** — *ETH Zürich - Department of Physics*

Supervisor Prof. Thomas H. Willwacher — *Sep 2026–Present*

- Investigating the structure of the Grothendieck–Teichmüller Lie algebra, with a focus on free Lie subalgebras and their generators.
- Studied operad theory, little-discs operads, braid groups, category theory, Hopf algebras, quantum groups, and Tannaka-Krein reconstruction, with applications to the Grothendieck-Teichmüller Lie algebra

**Temperature asymmetry in the Milky Way's hot circumgalactic medium induced by the Magellanic Clouds** — *University of Groningen - Kapteyn Astronomical Institute*

Supervisors Prof. Filippo Fraternali and Prof. Else Starkenburg — ***Published*** *Mar 2026*

- **Oprea, A.** et al. (2026). "Temperature asymmetry in the Milky Way's hot circumgalactic medium induced by the Magellanic Clouds." **Monthly Notices of the Royal Astronomical Society**, 547(4). DOI: 10.1093/mnras/stag319
- Analyzed the Large Magellanic Cloud's passage effect on hot gas around the Milky Way disk using an existing hydrodynamical particle simulation.

**Hierarchical Topology of the Cosmic Web Tidal Force Field (Bachelor's thesis)** — *University of Groningen - Kapteyn Astronomical Institute*

Primary supervisor Prof. Rien van de Weijgaert (Kapteyn Astronomical Institute), Second examiner Prof. Marcello Seri (Bernoulli Institute) — *Mar 2025–Jun 2025*

- Used persistent homology (Morse theory) and topological data analysis to characterize Cosmic Web structures in the Illustris-3-Dark simulation
- Introduced topological metrics—p-norms, persistence entropy, and Wasserstein distances—to measure stability and evolution

**Symmetries in the Wigner phase-space description of the one-dimensional Quantum Harmonic oscillator** — *University of Groningen - Van Swinderen Institute*

Supervisor Prof. Daniël Boer (Van Swinderen Institute) — *Feb 2024–Apr 2024*

- Modeled the one-dimensional quantum harmonic oscillator's phase space with Wigner quasiprobability distributions.
- Studied the phase-space action of the symplectic group associated with the Hamiltonian, isomorphic to $SU(1,1)$.

## Work Experience

**Fundamentals of Electronics** — *University of Groningen*

Teaching Lab Assistant — *Feb 2025–Mar 2025*

- Guided students in LTspice simulations of electronic circuits in computer labs.
- Led lab sessions for groups of up to 20, covering operational amplifiers and advanced electronics.
- Graded preparation work and reports from students.

**Calculus 1, Calculus 2, Electricity and Magnetism** — *University of Groningen*

Teaching Assistant — *Sep 2023–Feb 2025*

- Led two weekly tutorials, explaining course concepts and facilitating learning activities.
- Graded weekly homework, midterms, and final exams.
- Managed exam distribution and collection while answering students' questions.

## Selected Advanced Coursework

**University of Groningen / ETH Zürich** *Groningen & Zürich*

*2024–2026*

- **Mathematics:** Algebraic Topology I–II, Analysis on Manifolds, Functional Analysis, Probability and Measure, Commutative Algebra, Algebraic Geometry, Representation Theory, Lie Groups, Symplectic Geometry, Differential Geometry II
- **Theoretical Physics:** Quantum Field Theory, General Relativity, Conformal Field Theory, Integrable Systems, Symmetric Spaces

## Awards and Certificates

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| Jul 2026 | **Academic Scholarship,** Prahova Community Foundation | *Romania* |
| Nov 2023 | **KHMW Young Talent Incentive Awards,** Royal Holland Society of Sciences and Humanities | *Netherlands* |
| Sep 2023 | **Teaching Assistants Training,** Faculty of Science and Engineering, University of Groningen | *Netherlands* |
| Feb 2021 | **Mention,** The Mathematical Gazette Olympiad - National stage | *Romania* |
| Dec 2020 | **Harmonised Amateur Radio Examination Certificate (HAREC),** ANCOM Bucharest | *Romania* |
| Dec 2020 | **CEPT Novice Radio Amateur Licence (YO9OPR),** ANCOM Bucharest | *Romania* |
| May 2019 | **Mention and Silver Medal,** Physics Olympiad - National stage | *Romania* |
| Apr 2018 | **Mention and Silver Medal,** Physics Olympiad - National stage | *Romania* |

## Extracurricular activities

**Sirius A study association** *Groningen, Netherlands*

Active member of the Stargazing committee and the Starburst committee. *Sep 2022–Jul 2025*

- Organized an archery-tag event for first-year astronomy students (Starburst Committee).
- Organized a Stargazing Committee trip to the Eise Eisinga Planetarium, the world's oldest working planetarium.

**Astro Pi - European Space Agency** *Ploiesti, Romania*

*Sep 2020–Jan 2022*

- Proposed using Python and a Raspberry Pi aboard the International Space Station to detect potential anomalies in Earth's magnetic field.

## Skills

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| **Operating Systems** | Linux, MacOS, Windows |
| **Programming Languages** | C/C++, Python, Mathematica, C#, Java, Git |
| **Other Software Packages** | Inventor, Fusion360, Unity3D, Adobe Photoshop, LaTeX, BibTeX, LTSpice |
| **Soft Skills** | Time Management, Teamwork, Problem-solving, Leadership |

## Volunteer experience

**National Stargazing Days** *Groningen, Netherlands*

Kapteyn Institute *Feb 2023*

- Assembled, aligned, and focused a 300 mm Schmidt–Cassegrain telescope on the Bernoulliborg roof for public outreach.
- Gave short public talks on telescopes, astronomy, and the night sky.

**Nacht van de Nacht** *Groningen, Netherlands*

Kapteyn Institute *Oct 2022, Oct 2023*

- Assembled, aligned, and focused two 150 mm Schmidt–Cassegrain telescopes for public outreach.
- Gave short public talks on telescopes, astronomy, and the night sky.

**International Interact Club (district 2241)** *Ploiesti, Romania*

*Sep 2020–Jul 2021*

- Coordinated a career-orientation webinar series for 300 high-school students.
- Organized a Christmas fundraising recital with the Paul Constantinescu Philharmonic in Ploieşti.

## Languages

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|---|---|
| **English** | Professional proficiency - C1 certificate |
| **Romanian** | Native proficiency |
| **French** | Working proficiency |

**References available upon request.**